

Motor Power Analysis

2 matches

GACMP_E5 · 18 motors · 165.1s

GACOL_E8 · 18 motors · 165.1s

GACMP_E5 — Motor Output (Stator)

Motor	Max Voltage (V)	Avg Voltage (V)	Max Stator (A)	Avg Stator (A)	Peak Power (W)	Avg Power (W)	σ Power (W)	P95 Power (W)	Total Energy (Wh)
Accelerator	11.23	0.38	47.02	0.85	480.5	6.0	34.7	0.0	0.551
BL Drive	12.05	2.72	165.48	17.48	804.3	97.9	163.5	480.7	8.983
BR Drive	12.00	2.74	170.66	17.17	817.9	95.4	160.8	473.5	8.750
FL Drive	11.95	2.64	129.16	19.55	883.8	105.8	173.9	512.7	9.702
FR Drive	12.04	2.72	136.06	17.52	836.1	98.7	166.9	498.8	9.055
Flywheel 1	8.79	0.59	113.98	2.39	687.3	12.4	58.6	44.5	1.137
Flywheel 2	8.90	0.59	92.88	2.58	626.2	13.9	60.7	73.5	1.276
Flywheel 3	8.50	0.58	99.56	3.21	628.4	18.0	68.6	157.0	1.647
Hood	3.21	0.05	18.58	0.49	55.2	0.2	1.6	0.6	0.017
Indexer	10.79	0.34	63.16	2.04	540.5	14.7	73.1	0.0	1.347
Intake Extension 1	8.01	0.11	80.98	1.78	511.4	2.6	19.3	1.3	0.240
Intake Extension 2	21.53	0.10	82.56	1.57	295.0	2.2	16.8	2.1	0.198
Intake Roller 1	12.18	2.23	232.74	13.56	801.2	65.1	115.7	302.5	5.974
Intake Roller 2	13.17	2.23	148.32	13.58	812.3	65.1	111.7	292.8	5.972
BL Steer	9.45	0.55	64.42	5.01	425.7	8.7	27.4	49.6	0.798
BR Steer	9.55	0.55	64.64	5.08	400.6	8.5	25.6	48.1	0.779
FL Steer	8.55	0.54	66.06	5.30	417.1	8.7	26.8	47.8	0.794
FR Steer	7.60	0.59	75.74	5.59	454.7	10.4	32.0	59.6	0.950
TOTAL	0.00	0.00	0.00	0.00	2529.0	634.1	551.9	1644.2	58.169

GACOL_E8 — Motor Output (Stator)

Motor	Max Voltage (V)	Avg Voltage (V)	Max Stator (A)	Avg Stator (A)	Peak Power (W)	Avg Power (W)	σ Power (W)	P95 Power (W)	Total Energy (Wh)
Accelerator	12.99	0.93	75.68	1.46	662.5	12.5	55.4	77.4	1.149
BL Drive	12.01	1.83	75.76	7.34	492.1	33.1	66.4	195.0	3.037
BR Drive	12.11	1.88	85.08	7.63	476.7	34.8	68.6	203.5	3.189
FL Drive	12.12	1.54	92.16	8.82	526.2	37.1	73.5	214.2	3.404
FR Drive	12.09	1.69	72.88	8.45	512.2	37.9	73.4	216.1	3.480
Flywheel 1	11.24	1.12	135.02	2.96	1101.6	20.2	103.5	44.1	1.852
Flywheel 2	11.27	1.12	121.88	3.08	1117.2	21.2	106.6	49.8	1.947
Hood	10.27	0.15	23.38	0.69	139.1	0.3	1.7	0.8	0.025
Indexer 1	11.76	0.85	108.94	2.07	1045.8	16.6	63.9	190.2	1.527
Indexer 2	11.68	0.84	107.92	3.13	873.9	25.8	107.2	189.8	2.366
Intake Extension 1	10.42	0.26	74.50	3.99	561.0	4.5	25.2	11.1	0.411
Intake Roller 1	8.74	2.56	45.58	7.60	324.9	40.8	59.5	160.5	3.745
Intake Roller 2	8.71	2.56	42.56	7.09	311.3	38.0	55.5	148.6	3.483
BL Steer	6.45	0.37	62.78	2.79	355.3	3.6	17.8	13.7	0.330
BR Steer	6.67	0.37	69.44	2.88	316.9	3.8	16.1	14.2	0.345
FL Steer	6.86	0.39	65.66	3.15	388.8	4.0	17.6	17.8	0.368
FR Steer	7.73	0.41	64.02	3.05	480.8	4.5	20.0	19.5	0.409
TOTAL	0.00	0.00	0.00	0.00	2554.1	338.7	365.0	1059.1	31.064

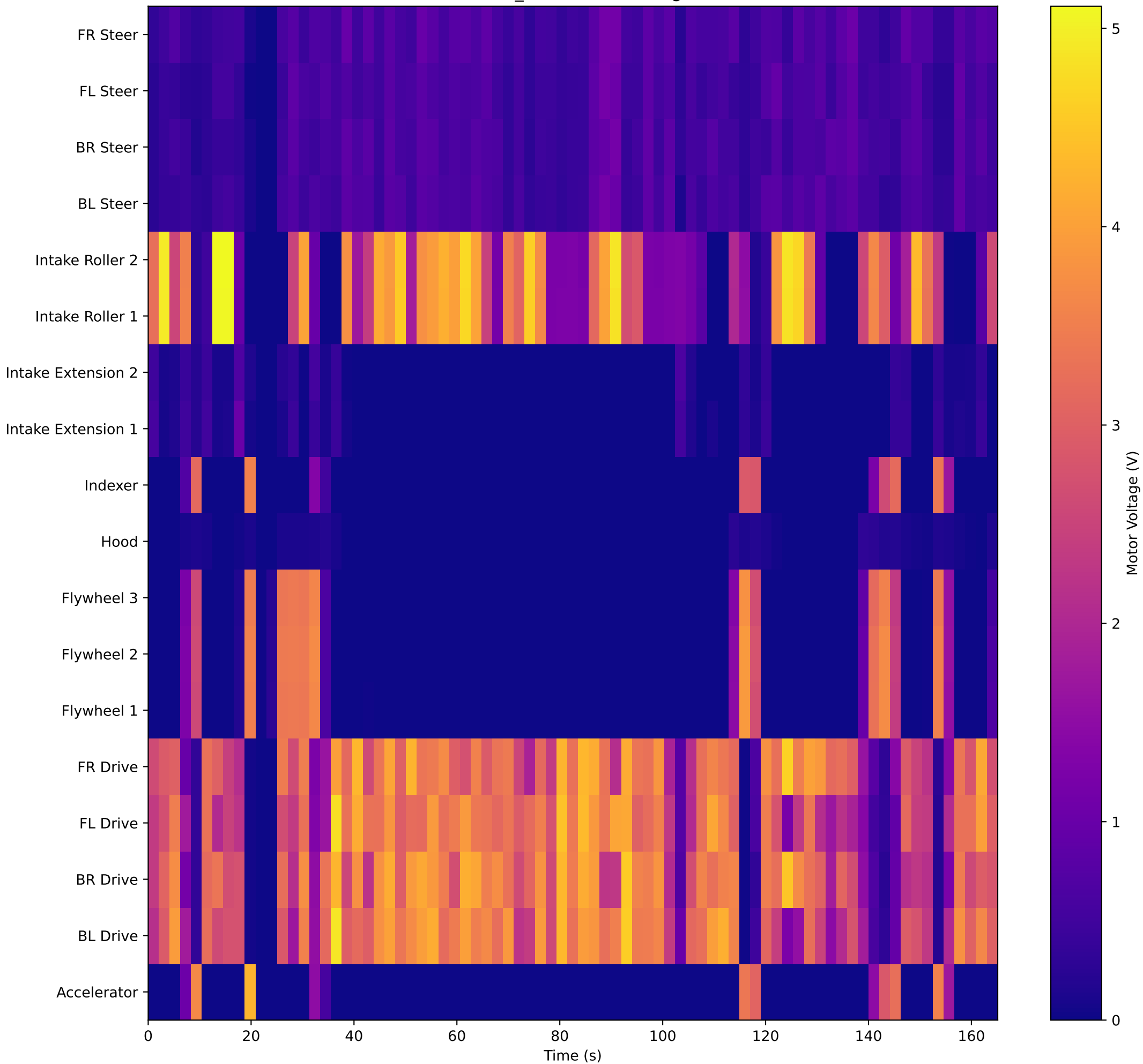
GACMP_E5 — Supply (Battery Draw)

Motor	Max Voltage (V)	Avg Voltage (V)	Max Current (A)	Avg Current (A)	Peak Power (W)	Avg Power (W)	σ Power (W)	P95 Power (W)	Total Energy (Wh)
Accelerator	12.85	4.85	37.30	0.65	382.3	6.1	33.6	0.0	0.558
BL Drive	13.82	4.87	84.20	10.77	804.1	97.9	163.5	480.4	8.980
BR Drive	13.80	4.89	83.30	10.48	817.1	95.3	160.8	473.4	8.747
FL Drive	13.88	4.87	88.32	11.70	884.0	105.7	173.9	512.5	9.699
FR Drive	13.85	4.88	87.24	10.89	835.3	98.7	166.9	498.7	9.053
Flywheel 1	12.85	4.86	48.42	1.42	450.1	12.7	55.9	57.8	1.169
Flywheel 2	12.85	4.86	43.10	1.62	429.4	14.4	58.7	98.1	1.322
Flywheel 3	13.55	4.83	48.02	2.15	392.1	18.4	66.9	196.0	1.689
Hood	12.88	4.86	6.96	0.03	51.5	0.3	2.5	0.5	0.025
Indexer	12.85	4.87	48.26	1.61	480.2	15.2	72.5	0.0	1.393
Intake Extension 1	12.85	4.85	20.38	0.29	260.9	3.0	19.8	1.4	0.273
Intake Extension 2	13.97	4.85	35.58	0.24	294.9	2.1	16.8	1.9	0.196
Intake Roller 1	12.88	4.60	69.12	7.33	512.8	63.0	104.8	287.6	5.784
Intake Roller 2	12.88	4.61	44.50	7.33	502.9	63.2	102.3	271.0	5.799
BL Steer	13.68	4.79	48.40	0.93	425.4	8.7	27.4	49.6	0.796
BR Steer	13.70	4.84	49.72	0.90	400.9	8.5	25.6	48.0	0.777
FL Steer	13.70	4.81	56.38	0.92	417.2	8.6	26.8	47.7	0.792
FR Steer	13.65	4.83	62.92	1.10	454.6	10.3	31.9	59.6	0.948
TOTAL	0.00	0.00	360.80	70.37	2527.5	632.2	542.2	1637.5	58.000

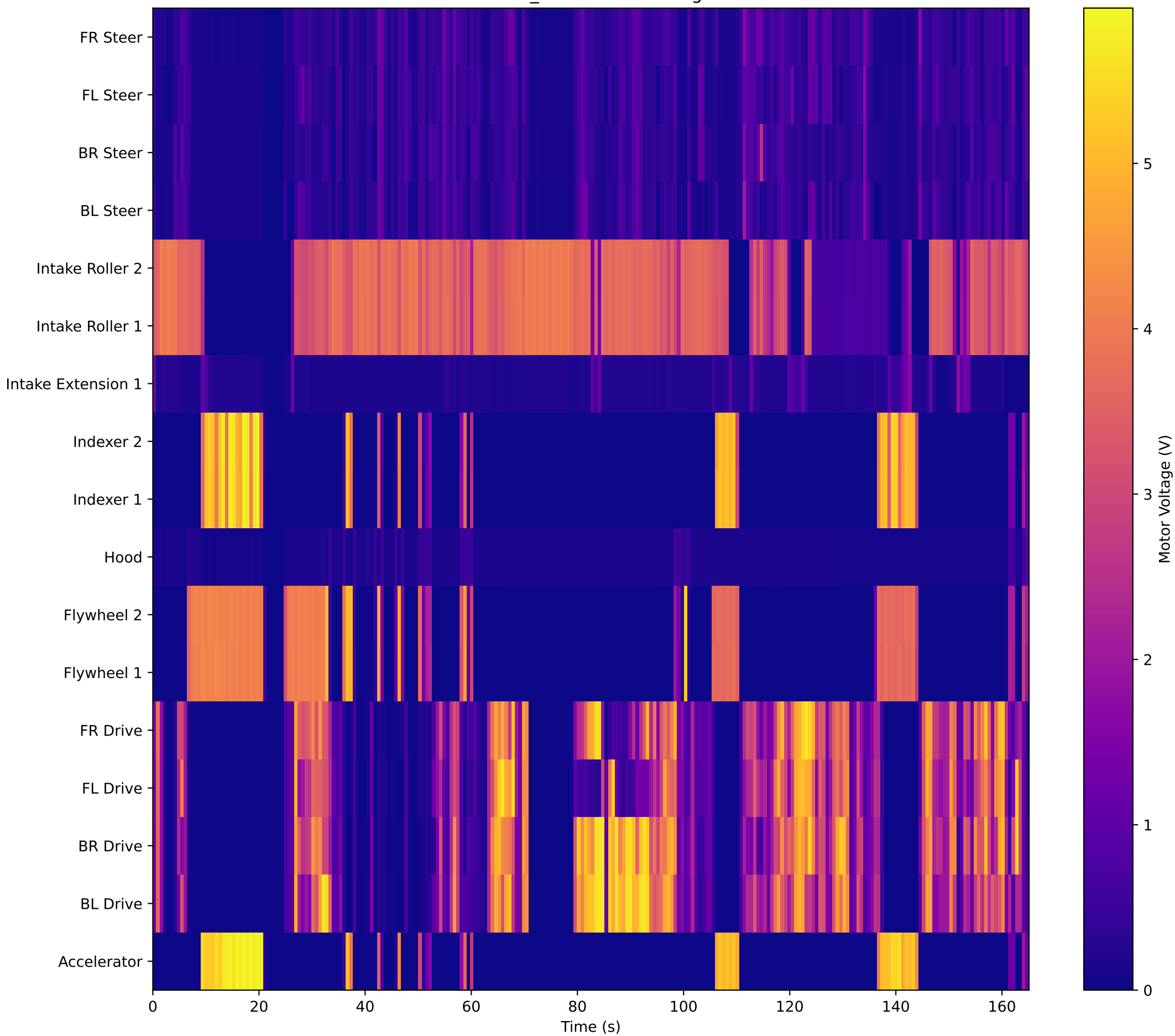
GACOL_E8 — Supply (Battery Draw)

Motor	Max Voltage (V)	Avg Voltage (V)	Max Current (A)	Avg Current (A)	Peak Power (W)	Avg Power (W)	σ Power (W)	P95 Power (W)	Total Energy (Wh)
Accelerator	13.12	5.62	60.80	1.27	557.8	13.2	55.3	80.3	1.211
BL Drive	14.38	5.65	46.18	3.04	491.8	33.1	66.3	195.0	3.034
BR Drive	14.43	5.65	42.46	3.19	476.6	34.7	68.6	203.4	3.187
FL Drive	14.43	5.65	45.56	3.40	526.2	37.1	73.4	213.8	3.402
FR Drive	14.40	5.66	47.82	3.46	511.7	37.9	73.4	216.1	3.477
Flywheel 1	13.12	5.62	73.58	1.91	750.7	19.7	92.2	51.9	1.807
Flywheel 2	13.15	5.63	72.16	1.93	743.7	20.1	90.9	55.0	1.842
Hood	13.15	5.62	12.04	0.03	158.0	0.4	4.8	0.7	0.035
Indexer 1	13.10	5.61	61.62	1.70	616.2	17.7	63.5	197.8	1.628
Indexer 2	13.15	5.61	73.64	2.70	756.8	27.7	107.4	226.9	2.537
Intake Extension 1	13.12	5.62	47.46	0.51	483.0	5.6	34.9	10.8	0.512
Intake Roller 1	13.05	5.54	25.68	3.75	275.9	40.7	57.8	163.0	3.732
Intake Roller 2	13.05	5.55	24.62	3.47	253.9	37.8	53.8	145.1	3.466
BL Steer	14.30	5.56	33.38	0.33	355.5	3.6	17.8	13.7	0.327
BR Steer	14.32	5.60	30.86	0.34	317.1	3.7	16.0	14.0	0.341
FL Steer	14.32	5.58	35.78	0.36	388.2	4.0	17.6	17.8	0.365
FR Steer	14.28	5.58	43.16	0.41	480.5	4.4	20.0	19.5	0.406
TOTAL	0.00	0.00	230.40	31.80	2102.8	341.3	352.1	1102.0	31.309

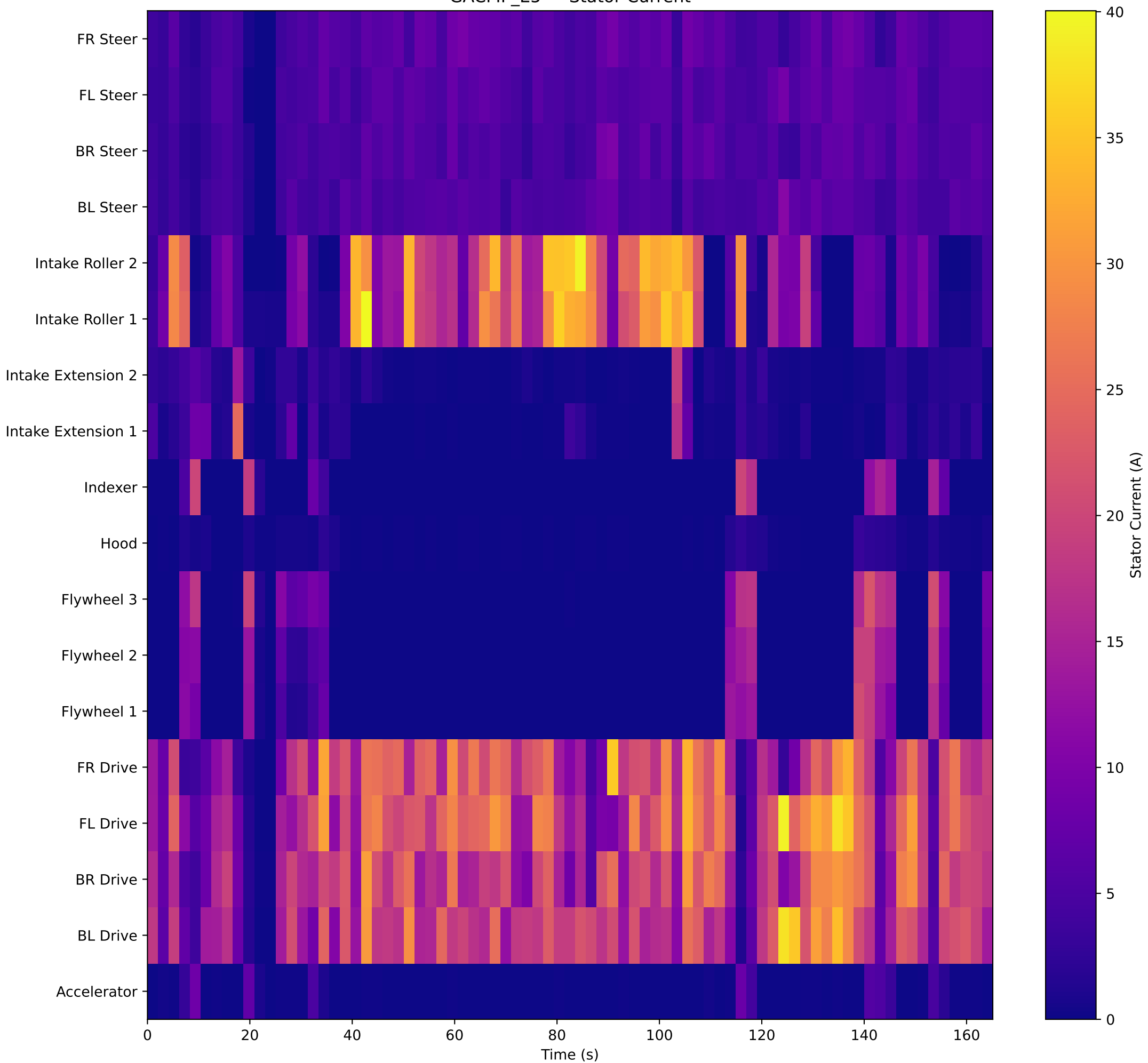
GACMP_E5 — Motor Voltage



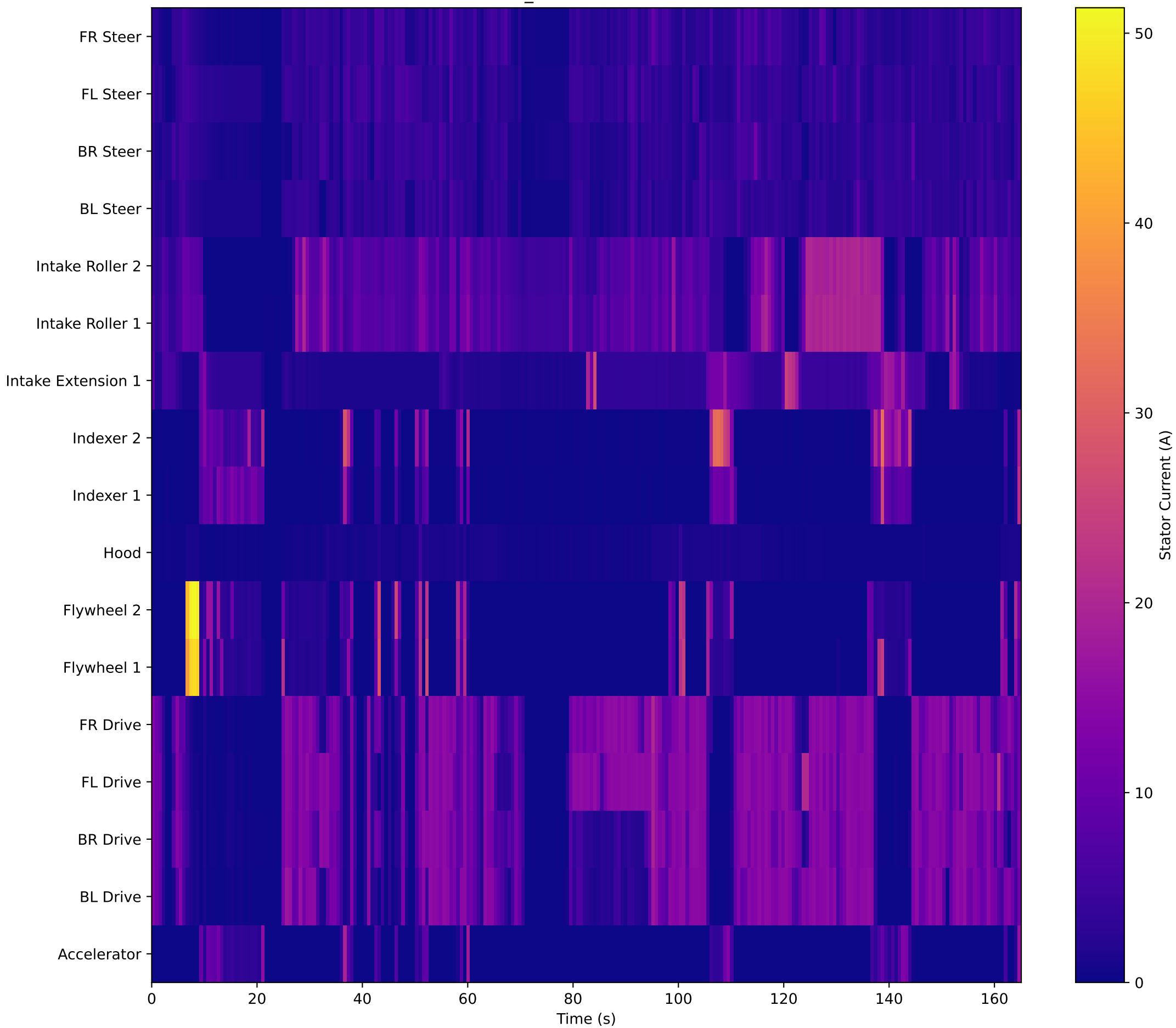
GACOL_E8 — Motor Voltage



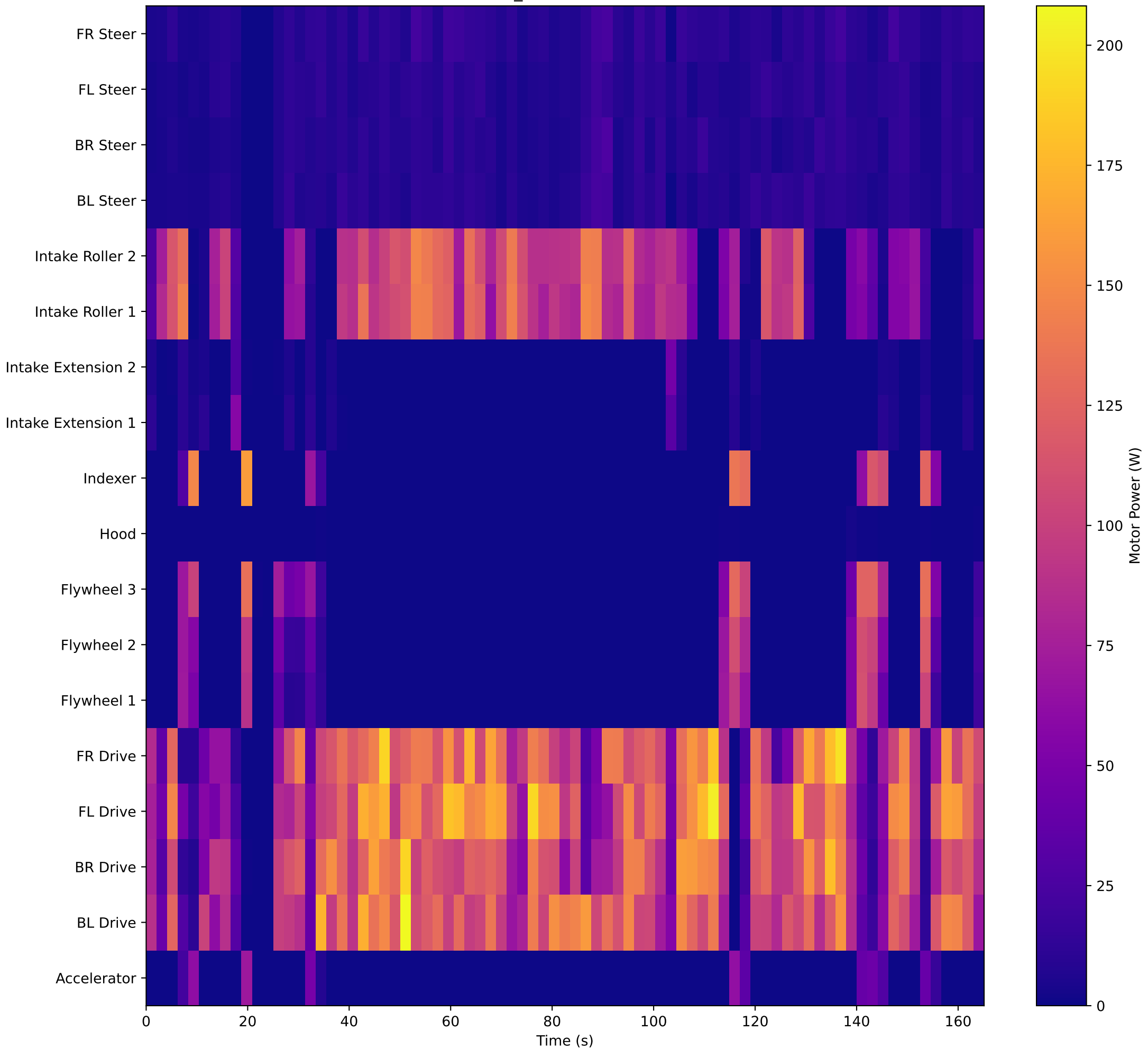
GACMP_E5 — Stator Current



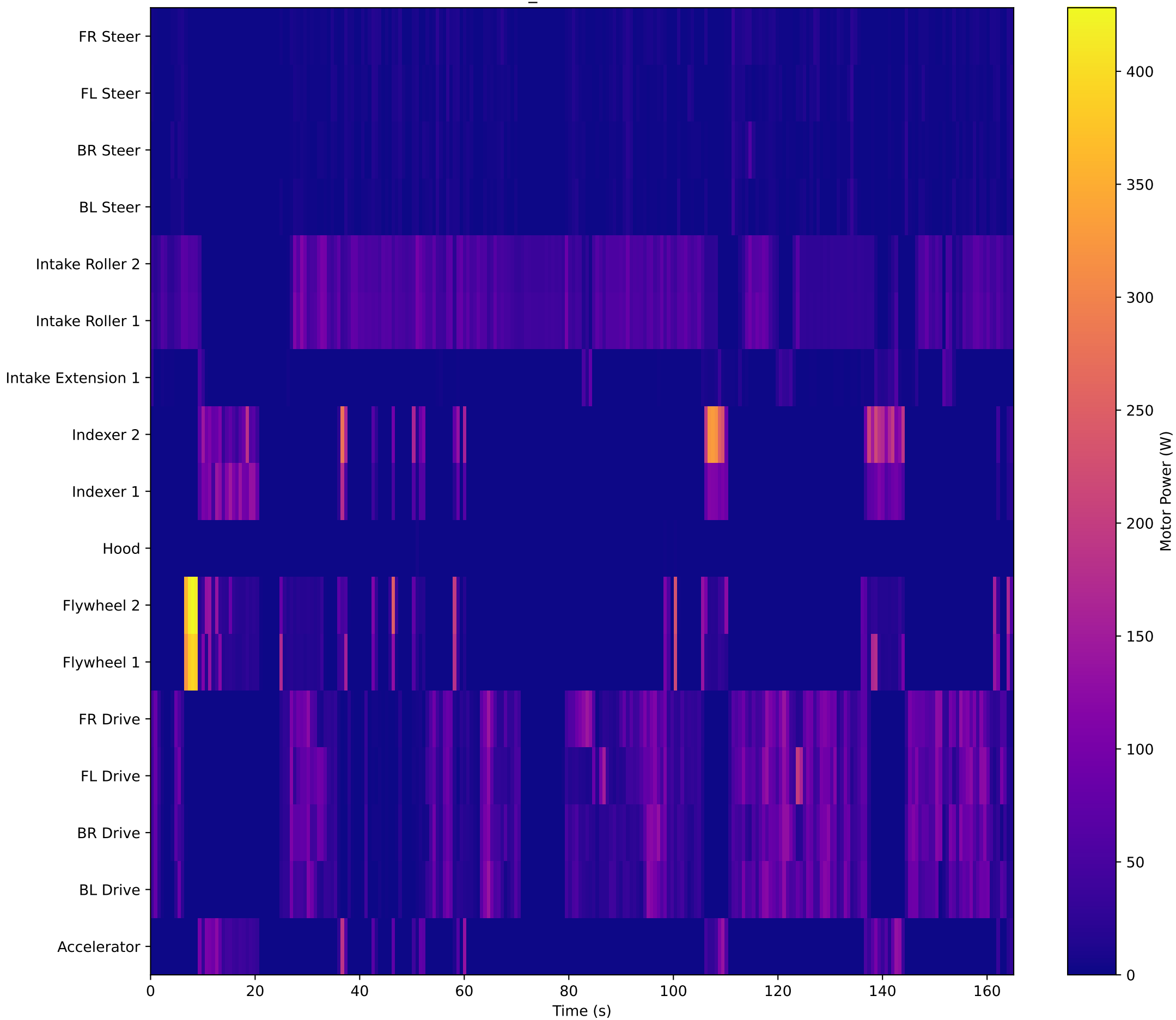
GACOL_E8 — Stator Current



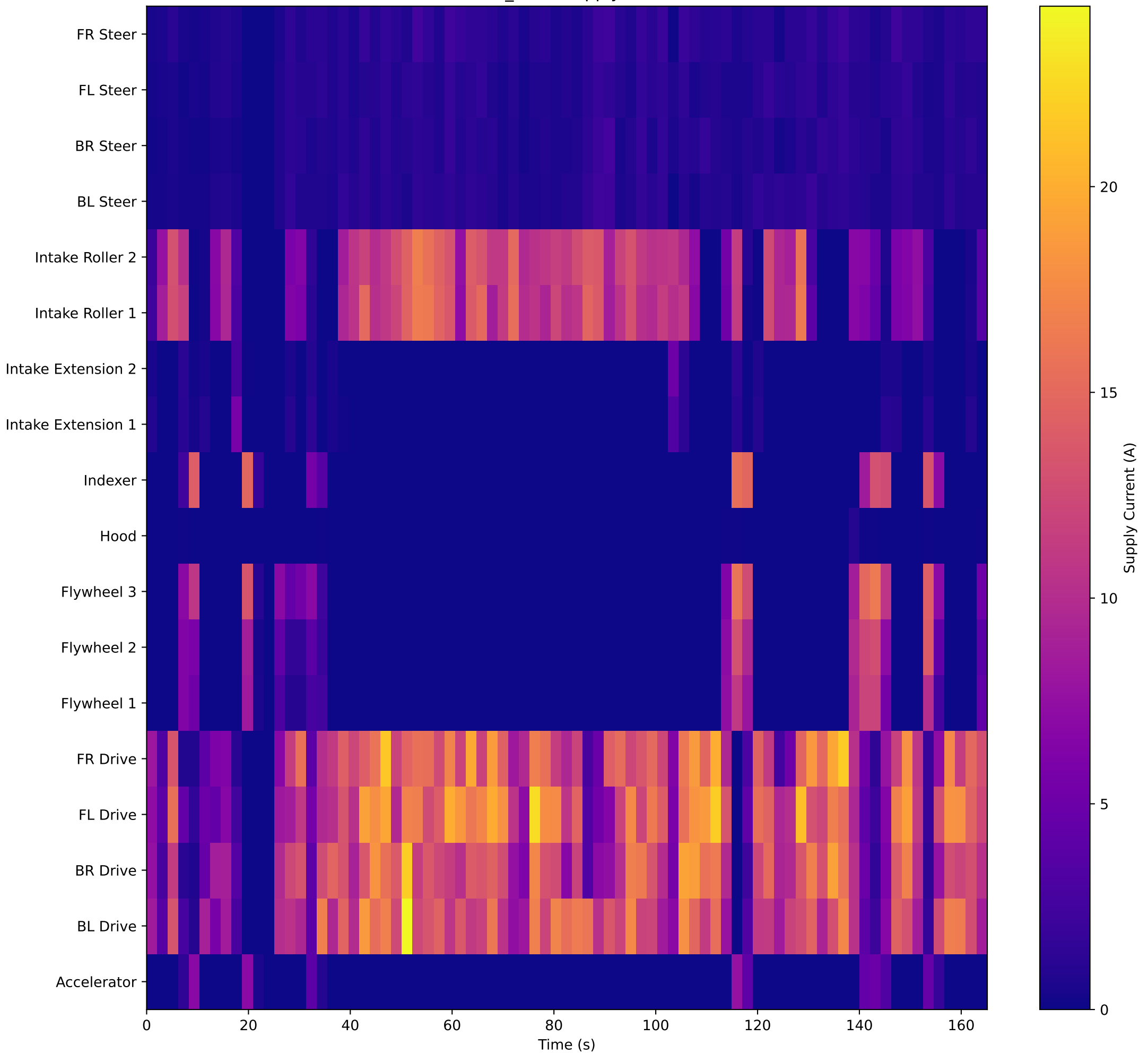
GACMP_E5 — Motor Power



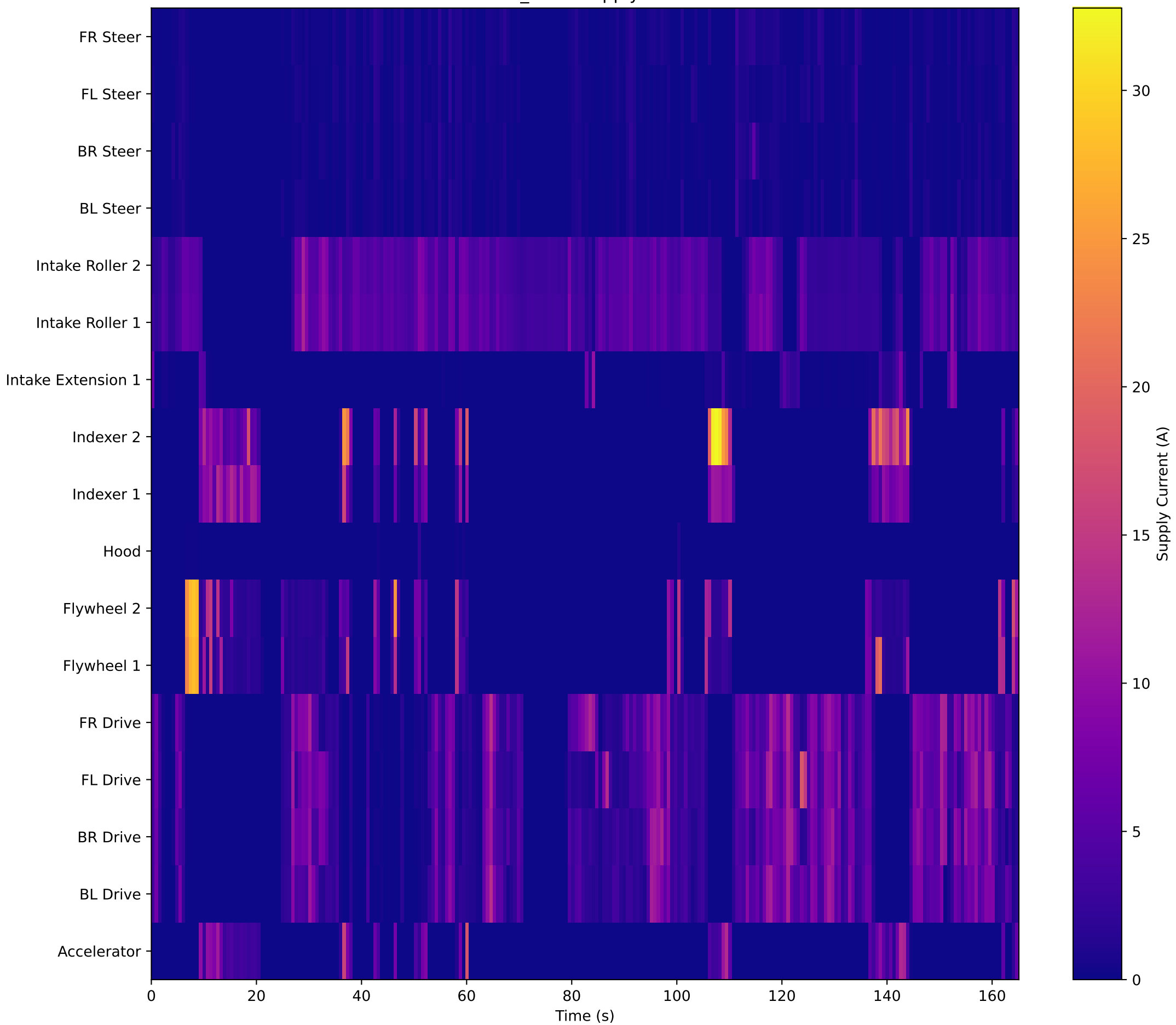
GACOL_E8 — Motor Power



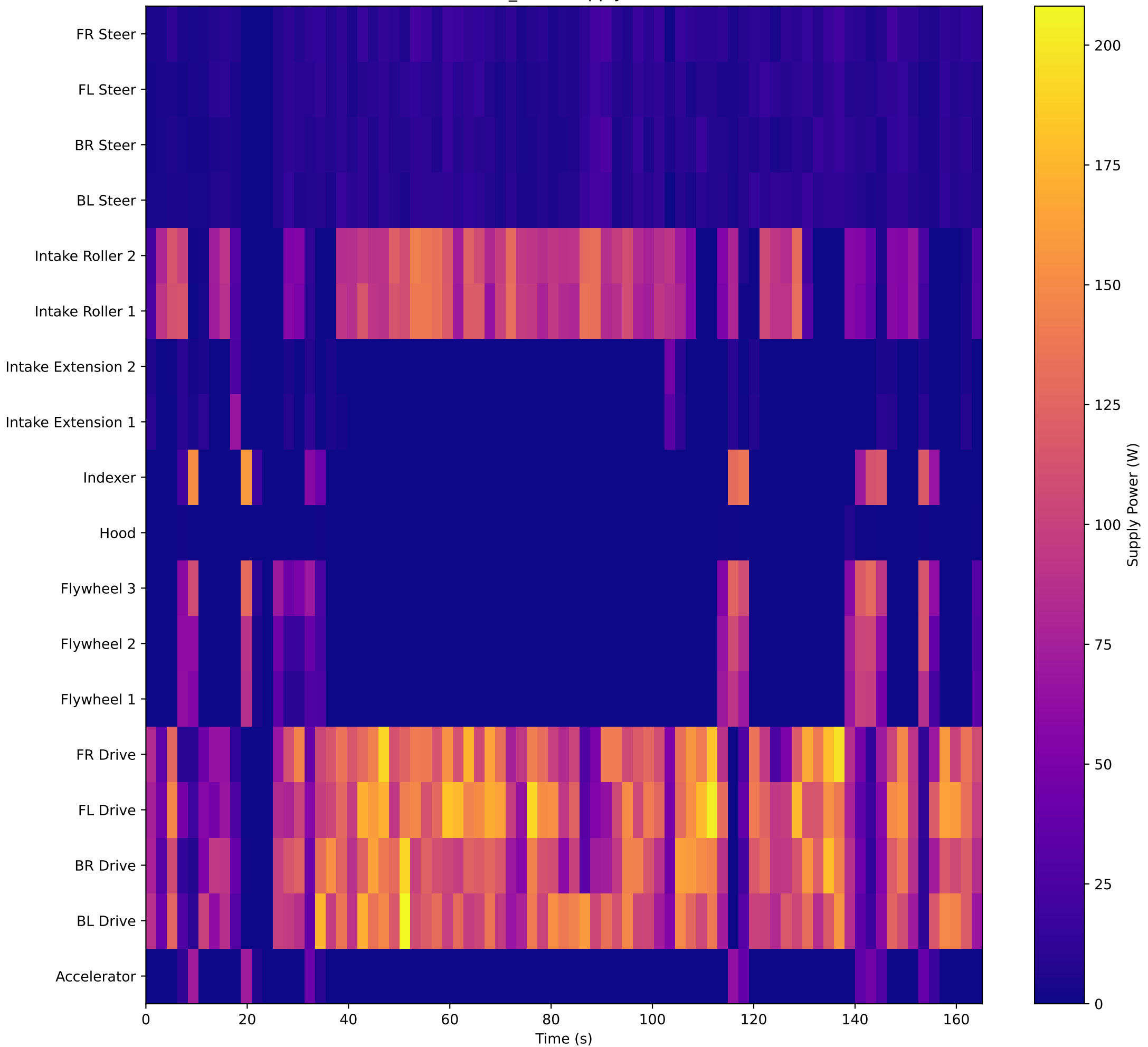
GACMP_E5 — Supply Current



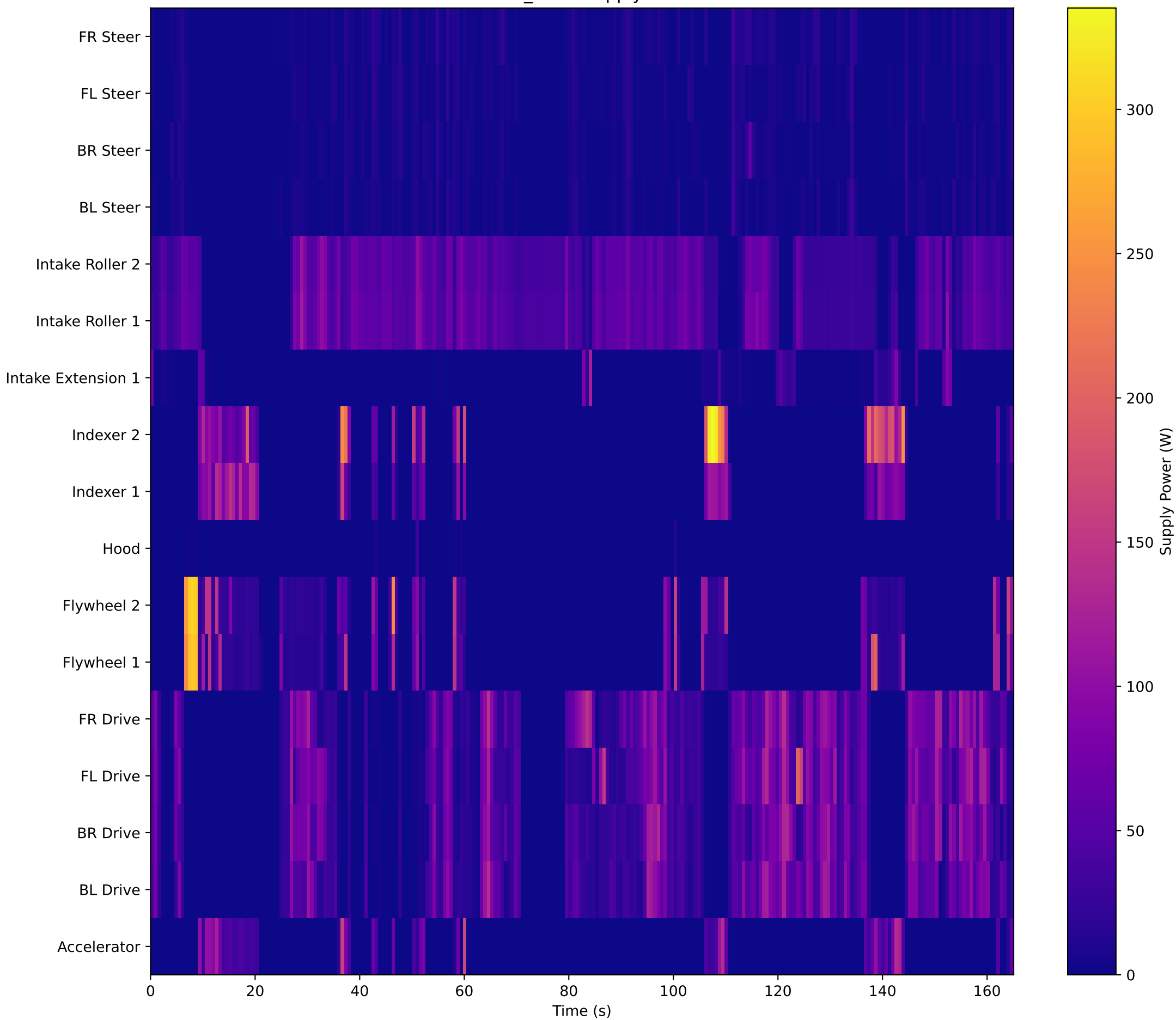
GACOL_E8 — Supply Current



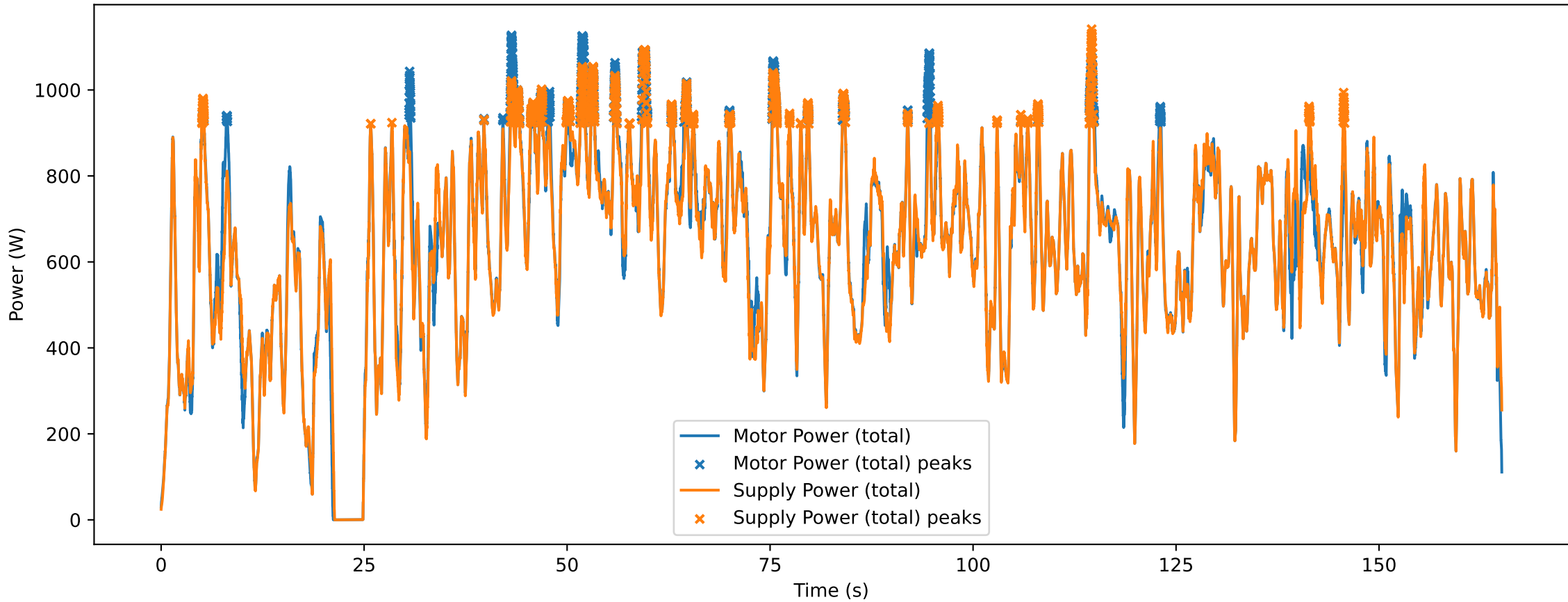
GACMP_E5 — Supply Power



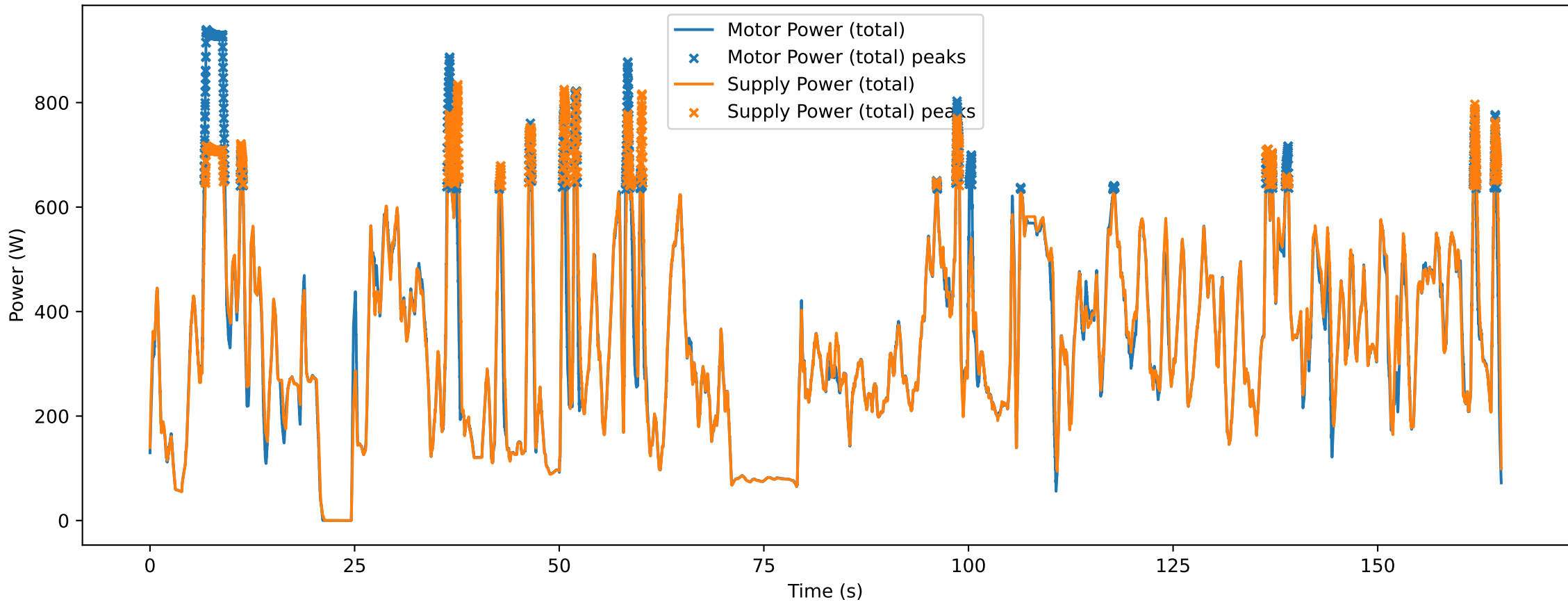
GACOL_E8 — Supply Power



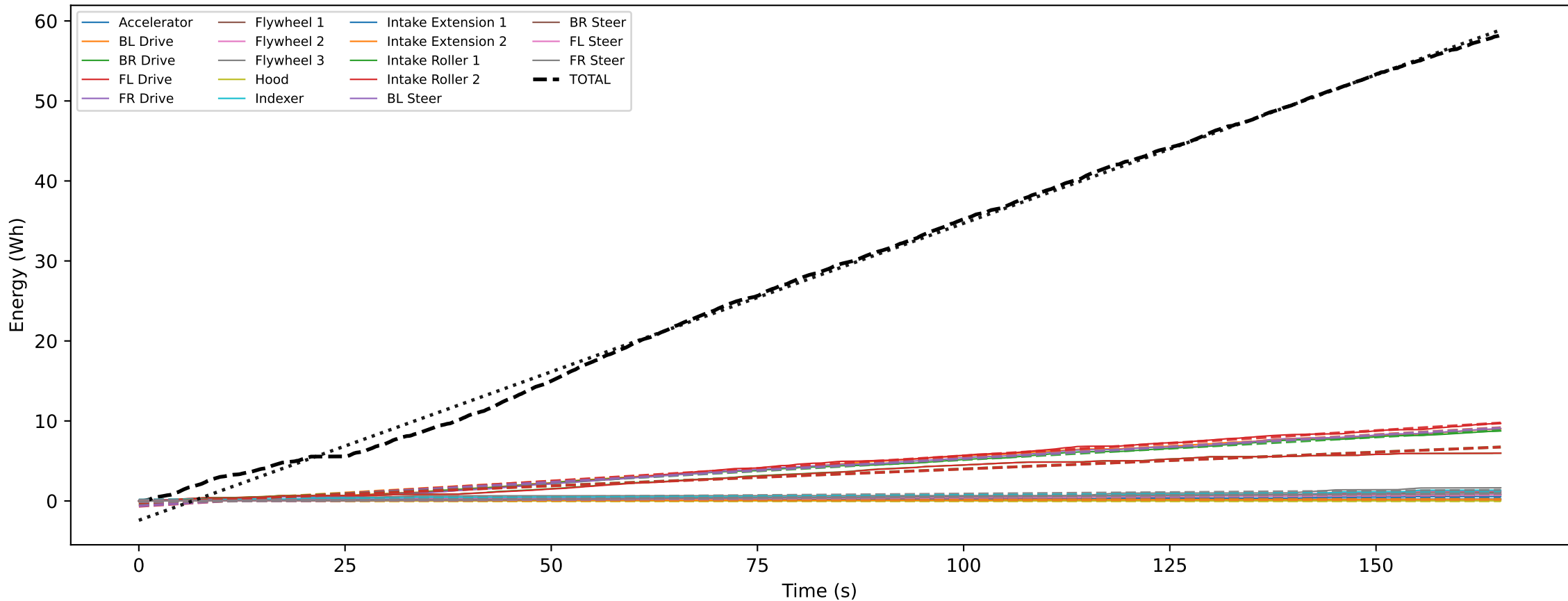
GACMP_E5 — Total Robot Power



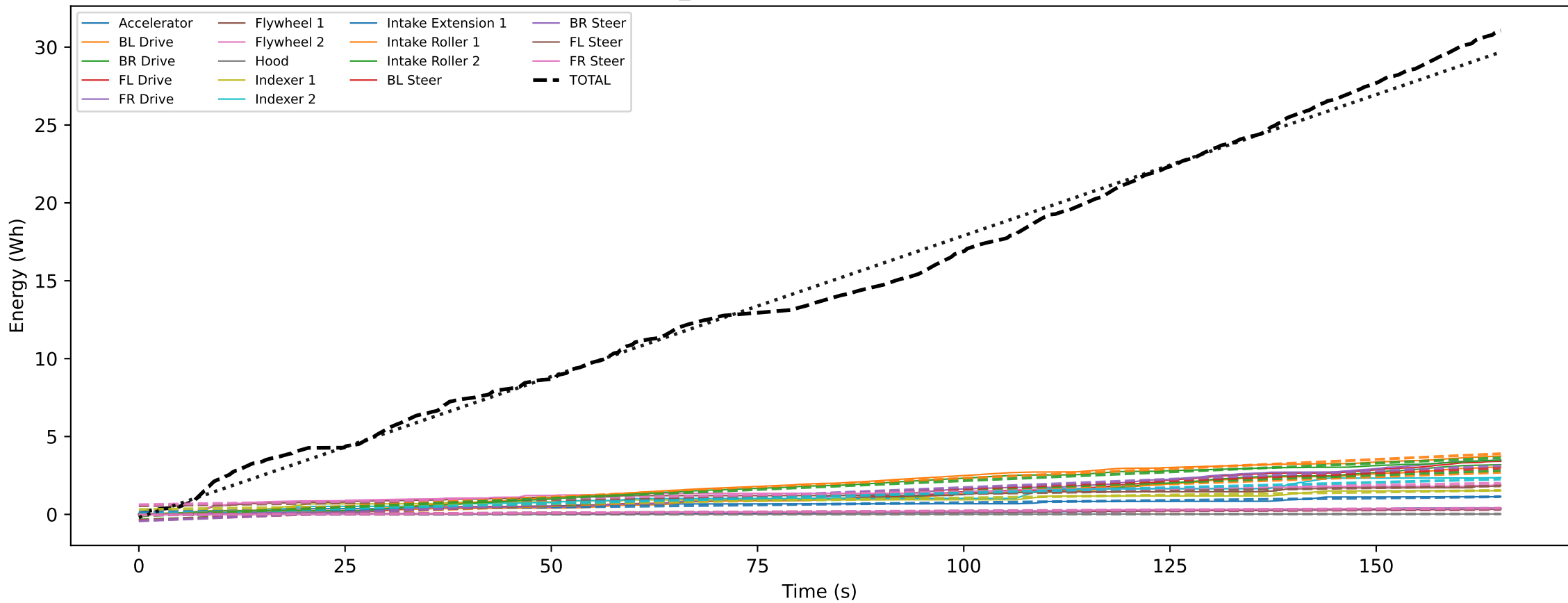
GACOL_E8 — Total Robot Power



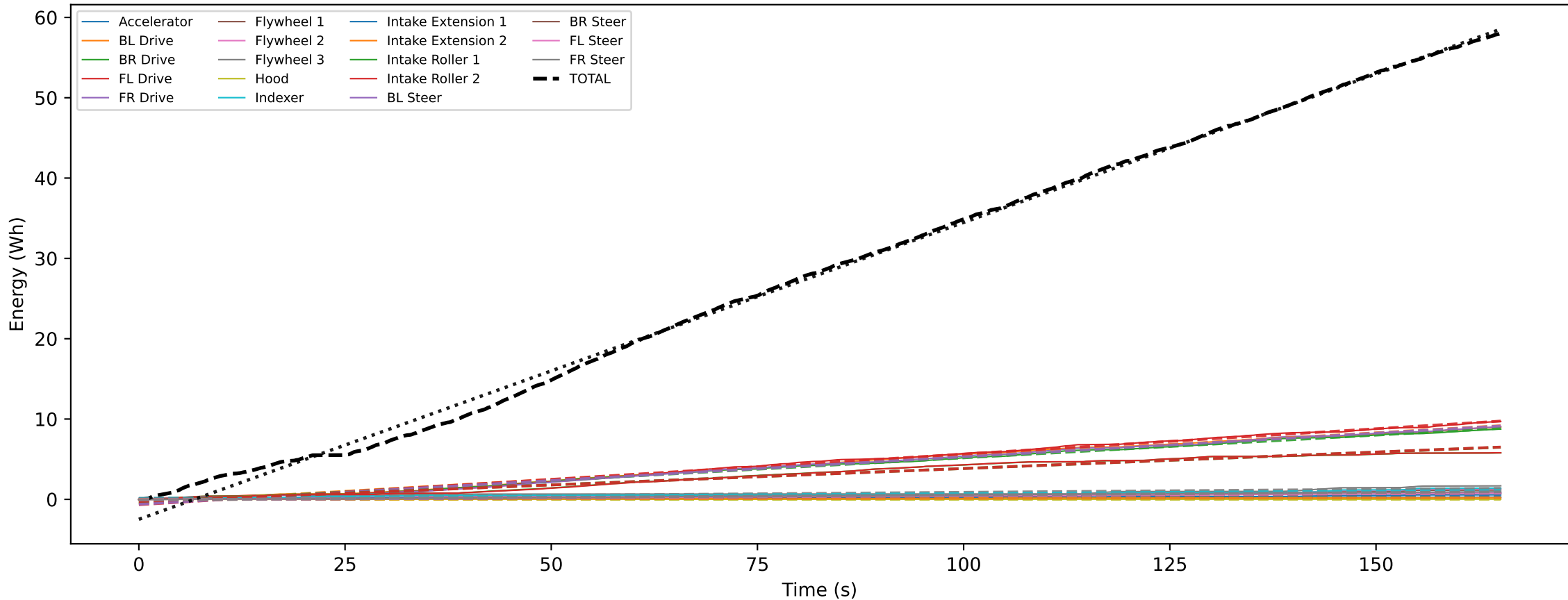
GACMP_E5 — Cumulative Motor Energy



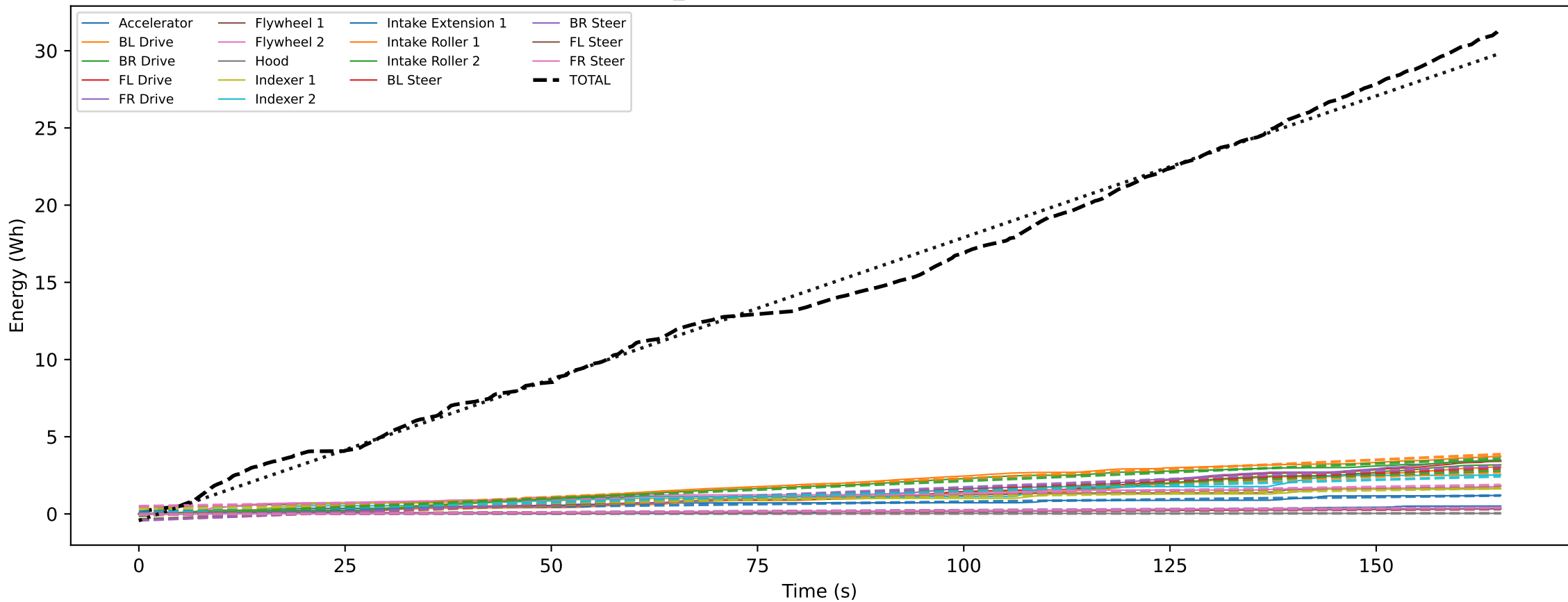
GACOL_E8 — Cumulative Motor Energy



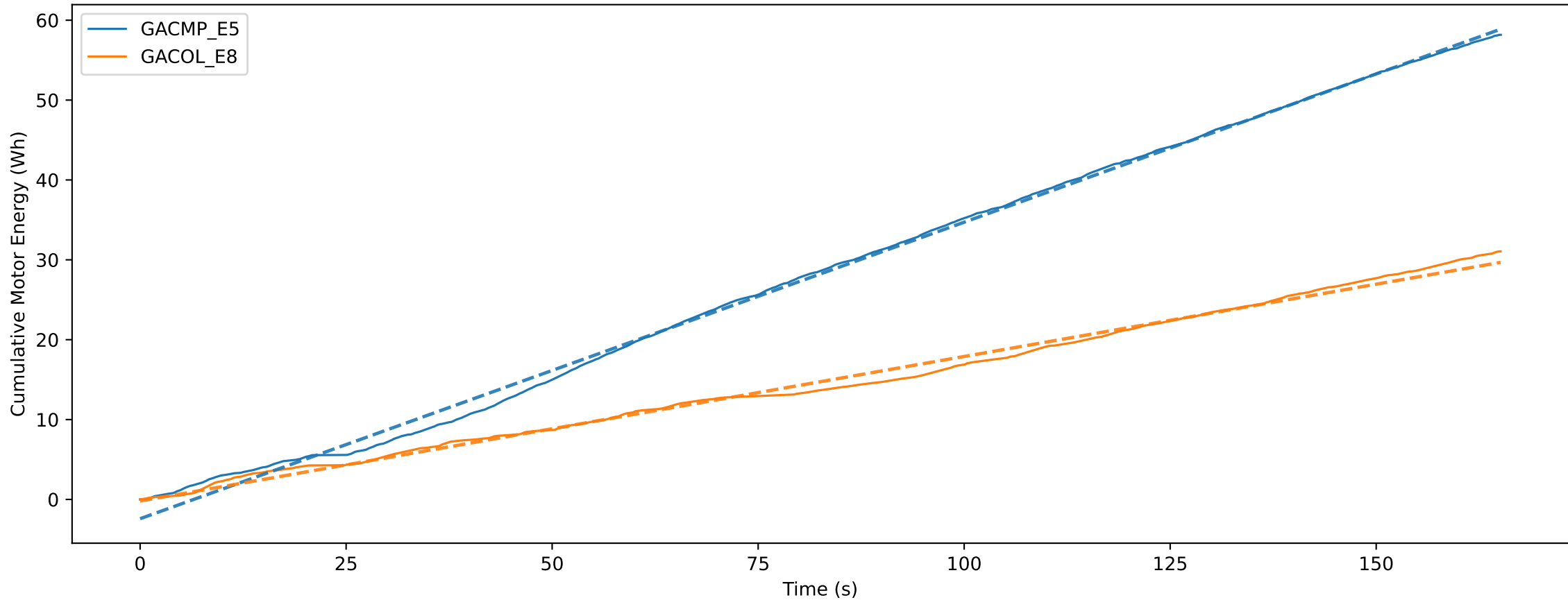
GACMP_E5 — Cumulative Supply Energy



GACOL_E8 — Cumulative Supply Energy



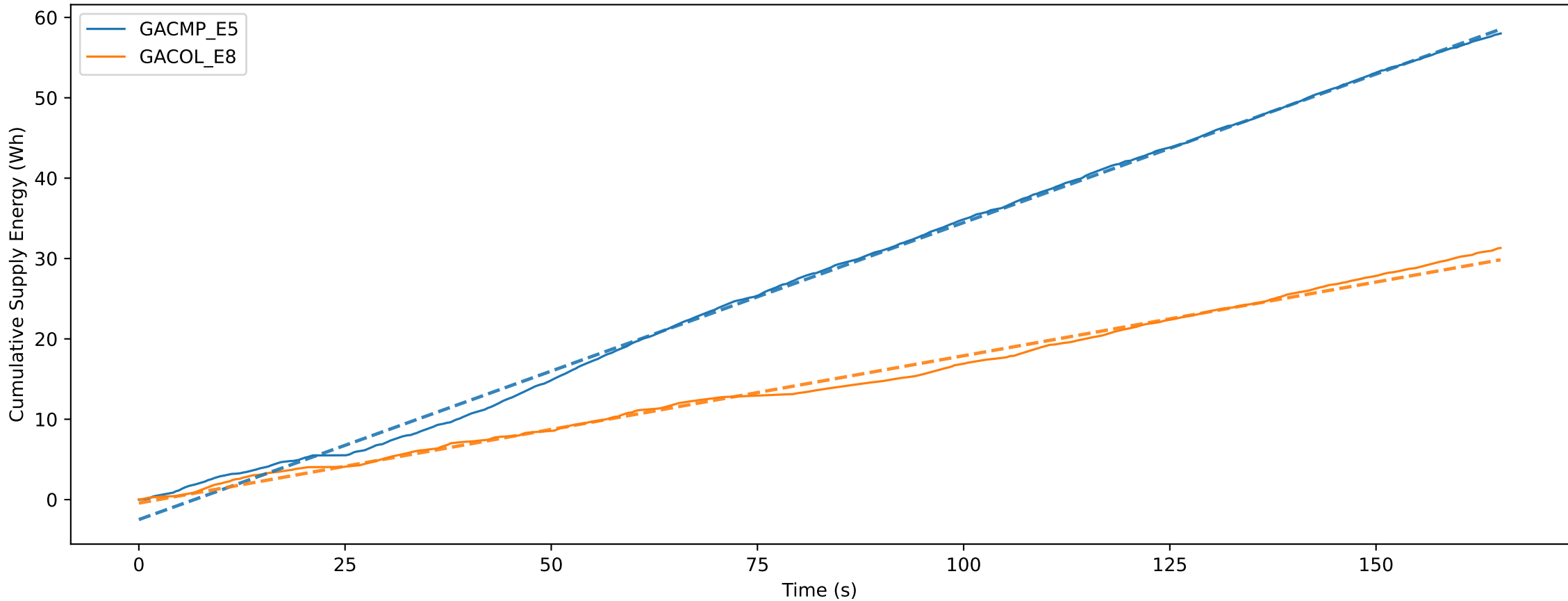
Match Comparison — Cumulative Motor Energy (Robot Total)



Match Comparison — Motor Energy (Wh)

Motor	GACMP_E5	GACOL_E8
Accelerator	0.551 Wh	1.527 Wh
BL Drive	8.983 Wh	3.037 Wh
BR Drive	8.750 Wh	3.189 Wh
FL Drive	9.702 Wh	3.404 Wh
FR Drive	9.055 Wh	3.480 Wh
Flywheel 1	1.137 Wh	1.852 Wh
Flywheel 2	1.276 Wh	1.947 Wh
Flywheel 3	1.647 Wh	0.000 Wh
Hood	0.017 Wh	1.149 Wh
Indexer	1.347 Wh	0.025 Wh
Indexer 2	N/A	2.366 Wh
Intake Extension 1	0.240 Wh	0.411 Wh
Intake Extension 2	0.198 Wh	N/A
Intake Roller 1	5.974 Wh	3.745 Wh
Intake Roller 2	5.972 Wh	3.483 Wh
BL Steer	0.798 Wh	0.330 Wh
BR Steer	0.779 Wh	0.345 Wh
FL Steer	0.794 Wh	0.368 Wh
FR Steer	0.950 Wh	0.409 Wh
TOTAL	58.169 Wh	31.064 Wh

Match Comparison — Cumulative Supply Energy (Robot Total)



Match Comparison — Supply Current (Robot Total)

